Daniel Engelberg

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Herzliya, Israel | Authorized to work in the US | Relocating to New York

SUMMARY

Robotics engineer (B.Sc., Technion) focused on navigation without GPS: built and benchmarked monocular visual odometry on drone data (6.5% drift per 10 m, 13 stress experiments), with coursework in kinematics, path planning, control and signal processing, and production Python experience.

SKILLS

Perception: OpenCV, visual odometry (KLT, essential matrix, RANSAC), camera calibration and frames, trajectory evaluation (ATE, RPE with evo), NumPy, SciPy
Robotics: ROS, kinematics, path planning, control theory, simulation, signal processing, RL (PPO)
Programming: Python, JavaScript, TypeScript, Node.js, SQL, MATLAB
CAD and analysis: SolidWorks, Onshape, Creo, AutoCAD, Kenesto; strength and thermal analysis; BOMs
Languages: English (native), Hebrew (fluent)

EDUCATION

Technion – Israel Institute of Technology, Haifa, Israel Mar 2026
B.Sc. in Mechanical Engineering, Robotics specialization
Relevant coursework: Robotics (Kinematics, Path Planning, Robotics Lab), Control Theory, Signal Processing, Python, CAD, Mechanical Design, Manufacturing Processes, Solid Mechanics, Fluid Mechanics

PROJECTS

Visual Odometry Drift Test | *Python, OpenCV, NumPy, evo* Oct 2026
- Built monocular visual odometry (KLT feature tracking, essential matrix with RANSAC, keyframing) and measured its drift against motion-capture ground truth on the EuRoC MAV drone dataset: 0.61 m error over an 80 m flight, 6.5% drift per 10 m
- Ran 13 controlled experiments (image noise, blur, darkness, dropped frames): noise more than doubled position error, while blur and dropped frames raised lost or rejected steps from 2 to 36 before accuracy changed
- Generated a synthetic downward-camera flight with exact ground truth and showed that flat terrain raises drift from about 3% to about 55% per 10 m (planar-scene ambiguity)
- Found and fixed two evaluation errors: a camera-versus-body frame mismatch with the IMU ground truth, and an OpenCV depth cutoff that rejected 2/3 of valid motion estimates

Hotel Robot Operating System
- Built a hotel operations system that dispatches delivery and logistics robots, coordinates elevator rides and takes requests from the property management system (PMS)
- Designed fault handling, persistent job state and audit trails covering elevator outages, duplicate jobs, invalid requests and failed deliveries

Handy AI | *Python, Flask, PostgreSQL, Claude / LLM APIs, WhatsApp Cloud API* Oct 2026 – Present
- Built a WhatsApp AI maintenance coordinator for property managers: tenants report issues by text, photo or voice note, and the system runs intake, collects contractor quotes, schedules visits and requests payment
- Wrote 350+ automated tests across 18,000+ lines of Python; deployed the web service and background worker on Render

EXPERIENCE

FlowTransit, Tel Aviv, Israel Jul 2026 – Sep 2026
Intern
- Studied 4 route-planning approaches (reinforcement learning with PPO, genetic algorithms, simulated annealing, shortest-path methods) to compare how each evaluates transit networks
- Improved GTFS and OpenStreetMap preprocessing so complete intracity bus lines are kept whenever any segment crosses a selected area, using map matching and network edge processing

Formula Student Technion, Haifa, Israel Oct 2023 – Apr 2024
High Voltage Engineer
- Designed the accumulator's high-voltage wiring architecture and thermal management, and ran strength analysis in SolidWorks
- Co-designed a new high-voltage accumulator for the race car from scratch in SolidWorks, managing files in Kenesto and tasks in monday.com